Scaling with GeoDNS

1. Overview

As the system grows globally (Asia, Europe, North America), a **single data center is no longer sufficient**.

Problem:

- High latency for distant users
- Backend overload → **504 Gateway Timeout errors**
- Thousands of requests per second

Solution:

> Deploy services in **multiple geographically distributed data centers** and route users intelligently using **GeoDNS**

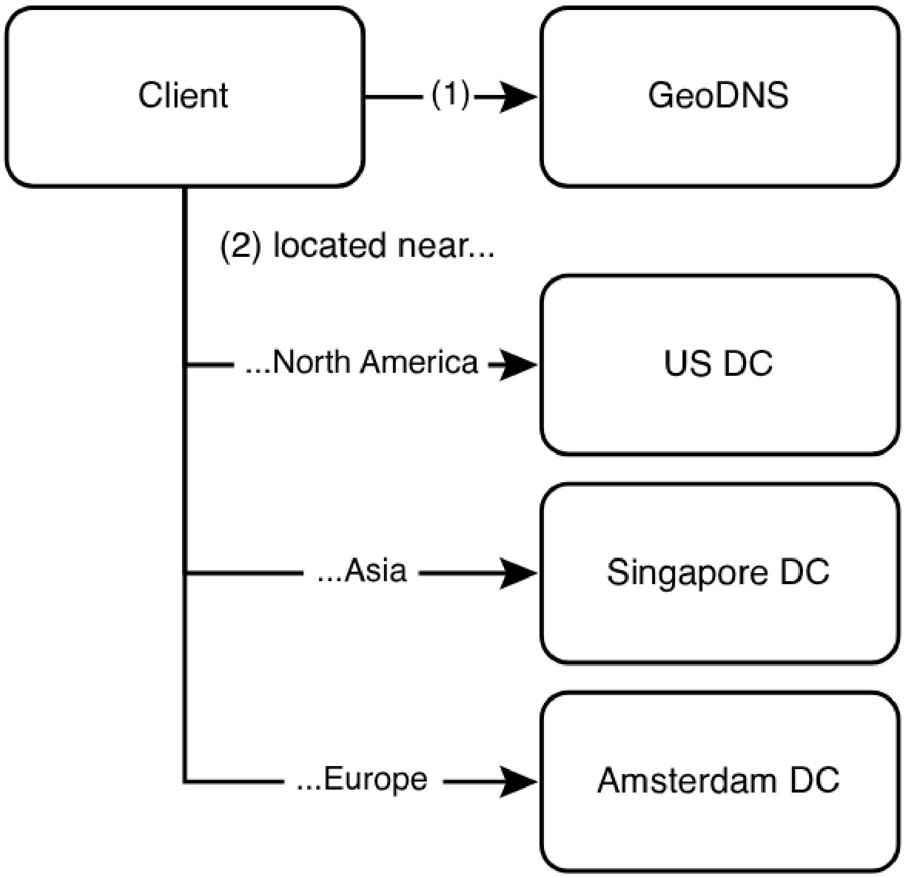
2. What is GeoDNS?

GeoDNS (Geolocation-based DNS) routes user requests to the **nearest data center** based on their geographic location (inferred via IP address).

Instead of returning a single IP:

- DNS returns **different IPs for different regions**

3. Architecture



4. Request Flow

1. User requests `beigel.com`
2. DNS query sent
3. GeoDNS detects user location via IP
4. Returns IP of **nearest data center**
5. Client sends request to that data center
6. Backend processes request

Client may cache this IP (important!)

5. DNS Configuration

- Multiple **A Records** for same domain:
    - US → IP1
    - Europe → IP2
    - Asia → IP3

GeoDNS decides which IP to return

6. Why GeoDNS is Needed

1. Reduce Latency

- Closer server → faster response

2. Improve Availability

- If one data center fails → route to another

3. Load Distribution

- Traffic split across regions

7. Key Requirement

Backend must be **stateless**

Why?

- Any request can go to any server
- No dependency on local session

8. Benefits

- Global scalability
- Low latency
- Fault tolerance
- Better user experience

9. Challenges

1. Data Consistency Across Regions

- Multiple DBs → sync issues

2. Replication Lag

- Writes in US may take time to reflect in Asia

3. GeoDNS Accuracy

- IP-based location is not always accurate

4. Failover Delay

- DNS caching may delay rerouting

10. DNS Caching

DNS responses are cached at:

- ISP
- OS
- Browser

Problem:

- If a data center fails, cached IP may still be used

Solution:

- Use **low TTL (Time-To-Live)**

11. Failover Strategy

If one data center is down:

- GeoDNS returns another region’s IP

BUT:

- Not instant due to DNS caching

12. GeoDNS vs Load Balancer

Feature	**GeoDNS**	**Load Balancer**
Level	Global	Within DC
Routing	Based on location	Based on load
Speed	Slower (DNS)	Faster
Use Case	Multi-region	Single region

Both are used together in real systems

13. Evolution from Previous System

Before:

- Single backend server

Now:

- Multiple backend servers
- Multiple data centers
- Geo-based routing

14. Final Summary

- GeoDNS routes users to **nearest data center**
- Uses **multiple A records**
- Reduces latency and improves availability
- Requires **stateless services**
- Limited by **DNS caching delays**